

Exam TFE4120

5th of June 2025

Solutions

Exercise 1: 10 marks (4, 4 and 2)

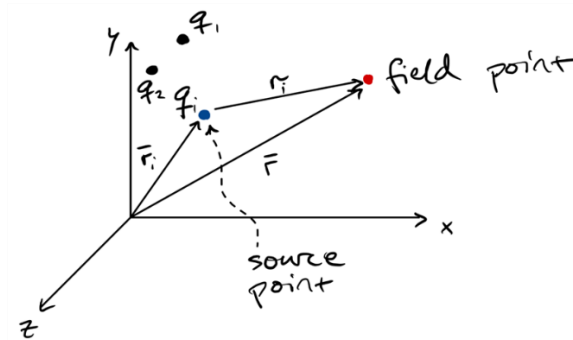
- Starting from Coulomb's law, prove that the total force from N stationary point charges on test charge Q is given by $\mathbf{F} = Q\mathbf{E}$, where $\mathbf{E}$ is the electric field from the source charges. Make a sketch of the situation and mark the charges and vectors you use in your proof.
- The total current crossing a surface S is $I = \int_S \mathbf{J} d\mathbf{a}_\perp = \int_S \mathbf{J} \cdot d\mathbf{a}$. Starting with this, derive the continuity equation. What is its physical meaning?
- Give the continuity equation in the case of magnetostatics and explain why it takes this form.

- Coulombs law is $\mathbf{F} = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2} \hat{\mathbf{r}}$ and gives the force on Q from q . For multiple charges, we have

$$\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \dots = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 Q}{r_1^2} \hat{\mathbf{r}}_1 + \frac{q_2 Q}{r_2^2} \hat{\mathbf{r}}_2 + \dots \right) = \frac{Q}{4\pi\epsilon_0} \left(\frac{q_1}{r_1^2} \hat{\mathbf{r}}_1 + \frac{q_2}{r_2^2} \hat{\mathbf{r}}_2 + \dots \right) = Q\mathbf{E}$$

where we have defined the electric field of the source charges as

$$\mathbf{E}(\mathbf{r}) \equiv \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i^2} \hat{\mathbf{r}}_i.$$



- We start with $I = \int_S \mathbf{J} \cdot d\mathbf{a}_\perp = \int_S \mathbf{J} \cdot d\mathbf{a}$ as given. When integrating over a closed surface S , this represents the charge per unit time leaving the volume $\mathcal{V}$ enclosed by the surface:

$$\oint_S \mathbf{J} \cdot d\mathbf{a} = \int_{\mathcal{V}} \nabla \cdot \mathbf{J} d\tau.$$

Charge is conserved, which means that charge flowing out from the volume means a decrease in charge left inside. This gives

$$\int_{\mathcal{V}} \nabla \cdot \mathbf{J} d\tau = - \int_{\mathcal{V}} \frac{\partial \rho}{\partial t} d\tau.$$

Since this holds for any volume, we get

$$\nabla \cdot \mathbf{J} = - \frac{\partial \rho}{\partial t}.$$

This is the continuity equation, and it describes local charge conservation.

- For magnetostatics, $\frac{\partial \rho}{\partial t} = 0$ since we only consider steady currents. The continuity equation in magnetostatics is thus $\nabla \cdot \mathbf{J} = 0$.

Exercise 2: 20 marks (10 and 10)

1. Three charges, one positive and two negative, of the same magnitude q are placed at three corners of a rectangle with sides a and $2a$ as shown in Figure 1 below. How much work is required to bring a fourth charge, $+q$, from far away and place it in the fourth corner of the rectangle as shown in the figure?

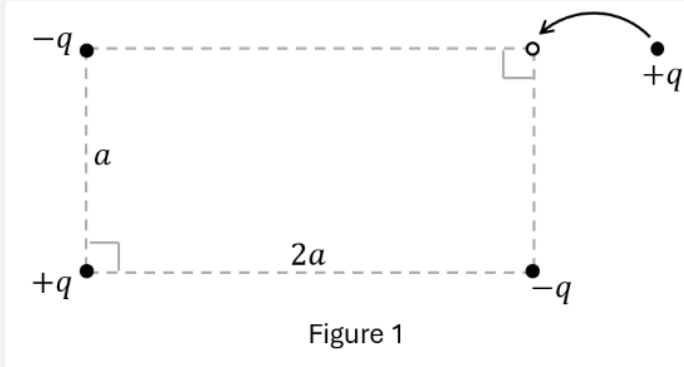


Figure 1

2. After the charge in task 1) has been placed, what is the work required to bring a fifth charge, $+q$, from far away and place it in the center of the rectangle as shown in Figure 2 below?

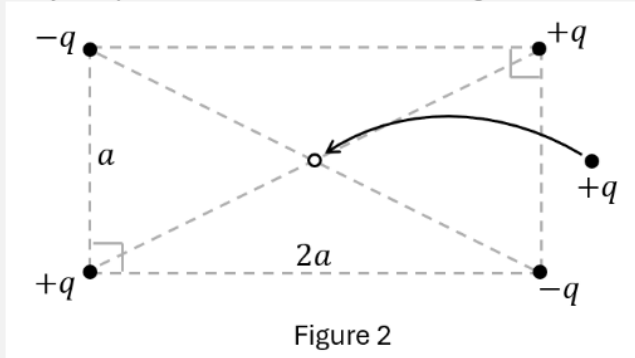


Figure 2

1. We start with calculating the potential $V = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_{ij}}$, since $W = qV$. We have

$$V = \frac{1}{4\pi\epsilon_0} \left\{ \frac{-q}{2a} + \frac{-q}{a} + \frac{q}{\sqrt{5}a} \right\} = \frac{q}{4\pi\epsilon_0 a} \left(\frac{1}{\sqrt{5}} - \frac{3}{2} \right).$$

The work required is therefore $W = +qV = \frac{q^2}{4\pi\epsilon_0 a} \left(\frac{1}{\sqrt{5}} - \frac{3}{2} \right).$

2. Similarly, we find the potential in the center of the rectangle

$$V = \frac{1}{4\pi\epsilon_0} \frac{2}{\sqrt{5}} \{ +2q - 2q \} = 0.$$

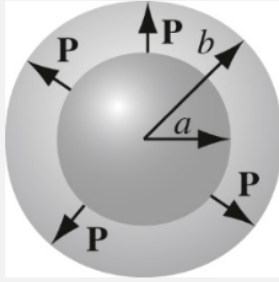
The work required in this case is therefore $W = +qV = 0.$

Exercise 3: 20 marks (10 and 10)

A thick spherical shell is made of an insulating material with a "frozen-in" polarization $\mathbf{P}(\mathbf{r}) = (1/r + k)\hat{\mathbf{r}}$, where k is a constant and r is the radial distance from the center. The shell has inner radius a and outer radius b as shown in the figure below. Find the electric field everywhere using the following methods:

1. Use the bound charges and Gauss's law.
2. Use Gauss's law for the electric displacement ($\mathbf{D}$), and use this and the polarization $\mathbf{P}$ to find the electric field.

Note that there is no free charge in the system.



1. We begin by finding the bound charges:

$$\rho_b = -\nabla \cdot \mathbf{P} = -\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \left(\frac{1}{r} + k \right) \right) = -\frac{1}{r^2} - \frac{2k}{r}$$

$$\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}} = \begin{cases} \left(\frac{1}{r} + k \right) \hat{\mathbf{r}} \cdot \hat{\mathbf{r}} = \frac{1}{b} + k & \text{at } r = b \\ \left(\frac{1}{r} + k \right) \hat{\mathbf{r}} \cdot -\hat{\mathbf{r}} = -\frac{1}{a} - k & \text{at } r = a \end{cases}$$

$$\text{Gauss's law } \oint \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{enc} \text{ gives } \mathbf{E} \cdot 4\pi r^2 = \frac{1}{\epsilon_0} Q_{enc} \Rightarrow \mathbf{E} = \frac{1}{4\pi r^2} \frac{Q_{enc}}{\epsilon_0} \hat{\mathbf{r}}.$$

We look at the enclosed charge and electric field in the three regions separately.

$$r < a: \quad Q_{enc} = 0 \Rightarrow \mathbf{E} = \mathbf{0}.$$

$$r > b: \quad Q_{enc} = 0 \Rightarrow \mathbf{E} = \mathbf{0}.$$

$$a < r < b: \quad Q_{enc} = \sigma_b|_{r=a} \cdot 4\pi a^2 + \int_a^r \rho_b \cdot 4\pi \tilde{r}^2 d\tilde{r} = \left(-\frac{1}{a} - k \right) 4\pi a^2 + \int_a^r \left(-\frac{1}{\tilde{r}^2} - \frac{2k}{\tilde{r}} \right) \cdot (4\pi \tilde{r}^2) d\tilde{r} = -4\pi a - 4\pi k a^2 + [-4\pi \tilde{r} - 4k\pi \tilde{r}^2]|_a^r = -4\pi r(1 + kr)$$

$$\Rightarrow \mathbf{E} = \frac{1}{4\pi r^2} \frac{1}{\epsilon_0} \cdot (-4\pi r(1 + kr)\hat{\mathbf{r}}) = -\frac{1}{\epsilon_0} \left(\frac{1}{r} + k \right) \hat{\mathbf{r}}.$$

2. We begin by finding the electric displacement $\mathbf{D}$ using Gauss's law for $\mathbf{D}$:

$\oint \mathbf{D} \cdot d\mathbf{a} = Q_{fenc} \Rightarrow \mathbf{D} = \mathbf{0}$ everywhere since there is no free charge in the system. Thus we have $\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P} = \mathbf{0} \Rightarrow \mathbf{E} = -\frac{1}{\epsilon_0} \mathbf{P}$. Again we look at each region separately:

$$r < a: \quad \mathbf{P} = \mathbf{0} \Rightarrow \mathbf{E} = \mathbf{0}.$$

$$r > b: \quad \mathbf{P} = \mathbf{0} \Rightarrow \mathbf{E} = \mathbf{0}.$$

$$a < r < b: \quad \mathbf{P} = \left(\frac{1}{r} + k \right) \hat{\mathbf{r}} \Rightarrow \mathbf{E} = -\frac{1}{\epsilon_0} \left(\frac{1}{r} + k \right) \hat{\mathbf{r}}.$$

Exercise 4: 20 (5+15) marks

1. Define the electrostatic energy and the total energy. What is the difference?
2. You have an insulating sphere of radius a that has relative permittivity ϵ_r . The sphere is uniformly charged with a total charge Q_f . Find the electrostatic and total energy of the system. Which of the two is greater?

1. Electrostatic energy is defined as $W = \frac{\epsilon_0}{2} \int E^2 d\tau$. It is the energy stored in the system. In other words, the work needed to bring both free and bound charge from infinity to the precise configuration. It does *not* include the work required to polarize a dielectric material. Total energy is defined as $W_{tot} = \frac{1}{2} \int \mathbf{D} \cdot \mathbf{E} d\tau$. It is the work needed to build up the free charge in the system from zero to ρ_f with the unpolarized dielectric in place, responding to the addition of free charge. The work needed to assemble the free charge depends on the bound charge in the dielectric.
2. We begin by finding $\mathbf{D}$ and $\mathbf{E}$ inside and outside the sphere. For simplicity we find only the magnitudes.

$$\begin{aligned}
 r > a: \quad Q_{f_{enc}} &= Q_f \\
 \oint \mathbf{D} \cdot d\mathbf{a} &= Q_{f_{enc}} \Rightarrow D \cdot 4\pi r^2 = Q_f \Rightarrow D = \frac{1}{4\pi r^2} Q_f \\
 E &= \frac{1}{\epsilon_0} D = \frac{1}{4\pi \epsilon_0 r^2} Q_f \\
 r < a: \quad Q_{f_{enc}} &= \frac{Q_f}{4\pi a^3/3} \cdot \frac{4\pi r^3}{3} = Q_f \cdot \frac{r^3}{a^3} \\
 \oint \mathbf{D} \cdot d\mathbf{a} &= Q_{f_{enc}} \Rightarrow D \cdot 4\pi r^2 = Q_f \cdot \frac{r^3}{a^3} \Rightarrow D = \frac{1}{4\pi} \frac{r}{a^3} Q_f \\
 E &= \frac{1}{\epsilon_r \epsilon_0} D = \frac{1}{4\pi \epsilon_r \epsilon_0} \frac{r}{a^3} Q_f
 \end{aligned}$$

The electrostatic energy is found by integrating the electric field squared, and splitting the integral between inside and outside the sphere:

$$\begin{aligned}
 W &= \frac{\epsilon_0}{2} \int E^2 d\tau = \frac{\epsilon_0}{2} \left[\frac{Q_f^2}{(4\pi \epsilon_r \epsilon_0 a^3)^2} \int_0^a r^2 4\pi r^2 dr + \frac{Q_f^2}{(4\pi \epsilon_0)^2} \int_a^\infty \frac{1}{r^4} \cdot 4\pi r^2 dr \right] \\
 &= \frac{Q_f^2}{8\pi \epsilon_0 a} \left(\frac{1}{5\epsilon_r^2} + 1 \right).
 \end{aligned}$$

The total energy is found by integrating the product of the electric displacement and the electric field, also here splitting the integral between inside and outside the sphere:

$$\begin{aligned}
 W_{tot} &= \frac{1}{2} \int \mathbf{D} \cdot \mathbf{E} d\tau = \frac{1}{2} \left[\frac{Q_f^2}{16\pi^2 a^6 \epsilon_r \epsilon_0} \int_0^a r^2 4\pi r^2 dr + \frac{Q_f^2}{16\pi^2 \epsilon_0} \int_a^\infty \frac{1}{r^4} \cdot 4\pi r^2 dr \right] \\
 &= \frac{Q_f^2}{8\pi \epsilon_0 a} \left(\frac{1}{5\epsilon_r} + 1 \right).
 \end{aligned}$$

We see that $W \propto \frac{1}{5\epsilon_r^2} + 1$ and $W_{tot} \propto \frac{1}{5\epsilon_r} + 1$. Since $\epsilon_r \geq 1$ for dielectric media and $\epsilon_r = 1$ only for vacuum, we have that $W_{tot} > W$.

Exercise 5: 10 marks (7 and 3)

1. We have a thick wire that has a circular cross-section of radius a . The wire carries a volume current density $\mathbf{J} = 3s^2$, where s is the radial distance from the wire axis. What is the total current of the wire?
2. The same wire now carries a uniform current $I = 3c^2$, where c is a constant. Find the volume current density $\mathbf{J}$.

1. The current density is non-uniform, so we need to integrate over the wire cross-section to find the total current:

$$I = \int \mathbf{J} \cdot d\mathbf{a} = \int_0^{2\pi} \int_0^a 3s^2 s ds d\phi = 6\pi \int_0^a s^3 ds = \frac{3\pi a^4}{2}.$$

2. The current is uniform, so we simply divide by the wire cross-section to find the volume current density:

$$J = \frac{dI}{da} = \frac{3c^2}{\pi a^2}.$$

Exercise 6: 20 (10+10) marks

1. Derive Ampère's law in magnetized materials, in other words that $\nabla \times \mathbf{H} = \mathbf{J}_f$. Start by defining the total current density $\mathbf{J}$ in a magnetized material and consider how the bound current $\mathbf{J}_b$ is related to the magnetization $\mathbf{M}$.
2. In the following two subtasks, consider linear magnetic media.
 1. Define how the magnetic susceptibility χ_m relates to the magnetization $\mathbf{M}$.
 2. Express the magnetic field $\mathbf{B}$ in terms of the auxiliary field $\mathbf{H}$, and use this to define the relative permeability μ_r .

1. The total current density is $\mathbf{J} = \mathbf{J}_b + \mathbf{J}_f$, and the bound volume current can be expressed as $\mathbf{J}_b = \nabla \times \mathbf{M}$. Ampère's law is $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$. Put together, this gives

$$\nabla \times \mathbf{B} = \mu_0 (\mathbf{J}_b + \mathbf{J}_f) = \mu_0 (\mathbf{J}_f + \nabla \times \mathbf{M}).$$

We collect the two curls and get

$$\nabla \times \left(\frac{1}{\mu_0} \mathbf{B} - \mathbf{M} \right) = \mathbf{J}_f.$$

We define the auxiliary field as $\mathbf{H} \equiv \frac{1}{\mu_0} \mathbf{B} - \mathbf{M}$, which gives $\nabla \times \mathbf{H} = \mathbf{J}_f$.

2. We have linear magnetic media.

1. $\mathbf{M} = \chi_m \mathbf{H}$

2. $\mathbf{B} = \mu_0 (\mathbf{H} + \mathbf{M}) = \mu_0 (\mathbf{H} + \chi_m \mathbf{H}) = \mu_0 (1 + \chi_m) \mathbf{H}$

We define the permeability $\mu \equiv \mu_0 (1 + \chi_m)$. The relative permeability is then

$$\mu_r = \frac{\mu}{\mu_0} = 1 + \chi_m.$$